

MD. FAIZUR RAHMAN KHAN

Senior Software Engineer

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Portfolio | LinkedIn | GitHub | HackerRank | LeetCode

PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 6 years of experience designing and building scalable production web applications end to end with **Next.js**, **React**, **TypeScript**, and **Node.js**, architecting database schemas and REST APIs on **PostgreSQL**, **MongoDB**, and **Redis**, and also comfortable with Express.js, NestJS, C#/.NET Core, and ASP.NET Core on the backend where a project calls for it. Deploys and operates on cloud platforms including **Vercel**, AWS (ECS, Lambda, EC2), and Google Cloud Platform, with containerized deployments using Docker and Kubernetes, and designs and maintains CI/CD pipelines with GitHub Actions and GitLab CI/CD. Comfortable owning a feature from database schema through API to UI, collaborating closely with designers on pixel-perfect UI implementation from Figma, delivering RESTful APIs, secure authentication with JWT and OAuth2, third-party integrations, performance optimization, and production reliability for SaaS products used by clients across international markets, including the US. Contributes to open-source projects on GitHub and practices Test-Driven Development. Proven people-side strengths: leading code reviews, mentoring and growing junior developers, and participating in sprint planning and client demos as part of cross-functional teams with frontend, QA, and DevOps.

CORE SKILLS

Backend: Node.js, Express.js, NestJS, .NET/C#, ASP.NET Core, RESTful & GraphQL API design, asynchronous & event-driven architecture, third-party integrations, microservices, service-oriented architecture

Frontend & Mobile: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Material-UI, D3.js, React Native, Vue.js, Angular, Redux, Figma (design handoff)

Languages: JavaScript, TypeScript, Python, C#

Databases & Caching: PostgreSQL (schema design & query optimization, pgvector), MongoDB, SQL Server, T-SQL, Entity Framework Core, Redis, Prisma, TypeORM, Mongoose

Cloud & DevOps: Docker, Kubernetes, Linux/Unix environments, AWS (ECS, Lambda, EKS, EC2, VPC, S3, RDS, OpenSearch, CloudFormation, CloudWatch), Vercel, Google Cloud Platform, Cloudflare

Search & Analytics: Elasticsearch (indexing, real-time search, aggregations), Kibana, Logstash

APIs & Integrations: REST, GraphQL, gRPC, OAuth2, JWT, SSO, Keycloak, Google/Facebook/Apple social login, Calendly, Airtable, Stripe, social-platform APIs (Pinterest, Instagram, Facebook, TikTok, LinkedIn, X, YouTube)

Async & Messaging: Bull/BullMQ, Redis queues, RabbitMQ, Hangfire, schedulers, retries/backoff, worker orchestration

Architecture & Business Systems: Distributed systems, microservices, service-oriented architecture, design patterns, data structures & algorithms, ticketing/booking platforms, ERP/CRM platforms, multi-tenant HR management systems, social-media management platforms

CI/CD, Quality & Observability: Git, GitHub Actions, GitLab CI/CD, Test-Driven Development (TDD), Jest, React Testing Library, Supertest, xUnit, ESLint, Prettier, Swagger, Postman, Prometheus, Grafana, ELK Stack

Security: Input validation, rate limiting, Helmet, CSRF protection, CORS hardening, route-level access guards, applied encryption & secure credential handling

AI-Assisted Development Tools: Claude Cowork, Claude Code, Codex, Copilot, Cursor, OpenClaw

AI/LLM Product Engineering: OpenAI, Claude, Gemini, RAG, prompt engineering, Python/FastAPI microservice, pgvector, Serper, Cartesia (TTS), Deepgram (STT), GPT Vision, query refinement, memory & context management, real-time voice/duplex mode

PROFESSIONAL EXPERIENCE

Chromatics AI LLC | Senior Software Engineer

Sep 2025 - Present

- Led the architecture and engineering of **Aisha.ai**, an **AI chat assistant** offered as a **SaaS product** and launched in the US, owning technical decisions across backend services, data, security, scalability, and production reliability.
- Architected and scaled Aisha.ai's distributed service-oriented **Node.js** backend using **NestJS**, **TypeScript**, **PostgreSQL**, **Prisma**, **Redis**, and **Docker**, separating core APIs, ingestion, and background workers for scalability and fault isolation.
- Built and integrated secure identity and access flows for Aisha.ai using **JWT** and **OAuth2/social login** with Google, Facebook, and Apple, plus **OTP verification**, password hashing and token **encryption**, password reset, token lifecycle management, and route-level access guards.
- Engineered background automation for Aisha.ai with **Bull/BullMQ** and **Redis** queues, including retries, exponential backoff, schedulers, worker orchestration, and **job observability** for scalable asynchronous workloads.
- Led production engineering practices for Aisha.ai across validation, **API contracts**, **security hardening**, automated testing, service boundaries, observability, and **Swagger** documentation while partnering with product and leadership on scalability, security, cost, and maintainability.
- Designed Aisha.ai's AI/LLM-integrated chat workflows (**OpenAI**, **Claude**, **Gemini**) and built a dedicated **Python/FastAPI** microservice for the **RAG** (retrieval-augmented generation) pipeline, handling embedding generation and semantic retrieval backed by **pgvector** in PostgreSQL.
- Integrated **Serper** for real-time web search, **Cartesia** for text-to-speech, and **Deepgram** for speech-to-text into Aisha.ai, including a **full-duplex voice mode** for real-time, natural back-and-forth conversations.
- Built **query refinement** and **memory/context management** pipelines for Aisha.ai to improve conversational continuity and the relevance and accuracy of AI responses across multi-turn chats.
- Developed file upload and analysis functionality using **GPT Vision**, enabling users to upload images and documents for **AI-powered analysis** and interpretation within Aisha.ai.

- Implemented **Markdown-formatted chat responses** in Aisha.ai, rendering headings, lists, bold/italic text, and code blocks for clearer, well-structured AI outputs.
- Implemented **end-to-end encryption** for user chat messages in Aisha.ai, protecting conversation content in transit and at rest.
- Built and maintained a social-media management SaaS product, on **ASP.NET Core**, **C#**, and **Entity Framework Core**, extending its multi-platform integration layer across Instagram, Facebook, TikTok, LinkedIn, X, YouTube, and Pinterest.
- Owned the **Pinterest** integration end to end: the OAuth2 connect flow, multi-board and multi-user account support, and pin publishing (image, video, carousel) and deletion via the Pinterest API, with **Hangfire**-scheduled background jobs for content publishing and analytics refresh.
- Diagnosed and fixed a data pipeline defect that silently blocked comment and reaction counts from populating after collection, and added on-demand engagement-metrics refresh backed by the Pinterest API.
- Redesigned publish/delete failure handling into a structured, machine-readable classification (authentication, permission, rate-limit, transient-service) surfaced through the API in place of a single generic error message, backed by new **xUnit** test coverage.

The Red IT | Senior Software Engineer (Remote)

Nov 2024 - Sep 2025

- Led development and maintenance of key **Node.js** features for the **Woztell CRM** and customer engagement SaaS/PaaS platform, supporting contact management, shared inbox workflows, WhatsApp customer interactions, and automation using **AWS Lambda**, **Express.js**, and **GraphQL**.
- Designed and delivered third-party SaaS integrations including **Calendly** and **Airtable** using **AWS Lambda** and APIs, extending automated WhatsApp workflows and platform capabilities.
- Planned and implemented a custom **Single Sign-On (SSO)** system for the inbox module, enabling secure cross-application authentication.
- Managed **Kubernetes**-based single-tenant PaaS deployments for client-specific isolated environments, ensuring security and customization.
- Improved platform scalability with separate read/write databases and enhanced inbox performance by **60%** through targeted refactoring and legacy-module optimization.

Sayburgh Solutions Limited | Senior Back End Developer

Feb 2022 - Oct 2024

- Led and mentored a team of **4 developers** while owning backend architecture and technical delivery across ticketing/booking, VOD streaming, and online education platforms, plus a multi-tenant ERP/HR management system.
- Built the backend for an **e-ticketing platform** for concerts and live band events, covering event and ticket inventory management and an end-to-end booking and payment flow for ticket purchases.
- Designed and optimized **MongoDB** and **PostgreSQL** schemas, indexes, joins, aggregation pipelines, and rate-limiting strategies for performance and system stability.
- Implemented **Redis** caching, **Docker** containerization, and **CI/CD** pipelines to streamline deployments and reduce release cycles.
- Built **multitenancy** support for the HR management system, enabling scalable onboarding of multiple organizations within a single deployment.
- Contributed to a **React Native** VOD mobile application alongside backend responsibilities, supporting cross-platform delivery and API integration.
- Contributed to an ERP-style restaurant management system using **Node.js microservices**, **gRPC** for inter-service communication, and **RabbitMQ** for asynchronous messaging.
- Integrated **Elasticsearch** for real-time search and analytics across the VOD streaming, HR management, and restaurant management platforms, improving search relevance and query performance over large datasets.
- Facilitated code reviews and mentored engineers to uphold software engineering best practices and maintain clean, testable architecture.

Jirle | Software Developer (Remote)

Sep 2022 - Dec 2023

- Developed and maintained a full-stack e-commerce platform using **Next.js** and **Tailwind CSS** for the frontend and **NestJS (Node.js)** for the backend.
- Delivered **RESTful APIs**, documented them using **Swagger**, and ensured high performance, security, and reliability.
- Refined **PostgreSQL** database design, improving query efficiency.
- Integrated **Elasticsearch** for product search and filtering on the e-commerce platform, improving search relevance and response times.
- Applied **Redis** caching and **Docker** for containerized deployments and participated in code reviews to maintain quality and consistency.

Brain Station 23 Limited | Associate Software Engineer

Jan 2021 - Sep 2021

- Contributed to three client projects, including **Robi Axiata Limited** and two international clients, delivering user-focused production features.
- Built responsive frontend applications using **React.js**, **Angular**, **Vue.js**, and **TypeScript** for refactoring and error handling.
- Implemented **Redux** store for efficient state management and improved frontend performance.

EDUCATION

B.Sc. in Computer Science & Engineering Daffodil International University	2015 - 2020
H.S.C. in Science Dhaka Megacity College	2012 - 2014
S.S.C. in Science Safiuddin Sarker Academy & College	2010 - 2012